

FedMed — Week 1 Development Plan

Centralized Baseline + Federated Learning Node Scaffolding

Week 1 Objective

Build the foundation of FedMed by first training a standard centralized 3D U-Net on a public brain MRI segmentation dataset such as BraTS, then setting up a Flower federated-learning environment with three mock hospital clients.

End-of-Week Deliverables

Deliverable	Expected Result
Centralized 3D U-Net	A working medical-image segmentation model.
Baseline metric	Validation Dice score and loss recorded.
Flower server	Federated server starts successfully.
3 mock hospitals	Hospital A, B, and C connect as independent clients.
First federated round	Clients receive/train/send model updates and the server aggregates them.

Day-by-Day Plan

Day 1 — Environment Setup

- Create the FedMed project and Python virtual environment.
- Install PyTorch, MONAI, Flower, NumPy, NiBabel, Matplotlib, and scikit-learn.
- Check whether CUDA/GPU acceleration is available.
- Create requirements.txt.

Day 2 — BraTS Dataset

- Obtain a public, de-identified BraTS dataset.
- Start with a small subset of approximately 5–10 subjects.
- Load one MRI volume and its segmentation mask.
- Verify image and label dimensions and visualize a few slices.

Day 3 — 3D U-Net

- Create a MONAI 3D U-Net.
- Set spatial_dims=3.
- Build the preprocessing pipeline.
- Pass one MRI volume through the network and verify tensor shapes.

Day 4 — Centralized Training

- Create the DataLoader.
- Use Dice loss or a Dice-based segmentation loss.
- Train the model for a small number of epochs.

- Confirm that the training loss decreases.

Day 5 — Baseline Experiment

- Run a repeatable centralized training experiment.
- Evaluate on a validation split.
- Record Dice score and validation loss.
- Save the baseline model checkpoint and experiment settings.

Day 6 — Flower Scaffolding

- Create a Flower server.
- Create three independent mock hospital clients.
- Start Hospital A, B, and C as separate processes.
- Verify that all three clients connect to the server.

Day 7 — First Federated Round

- Connect the 3D U-Net to the Flower clients.
- Give each hospital a different local data partition.
- Run one federated training round.
- Aggregate client model updates into a global model.
- Confirm that the global model changes after aggregation.

Recommended Week 1 Architecture

Centralized baseline: BraTS → preprocessing → 3D U-Net → centralized training → validation Dice score.

Federated scaffolding: Flower Server → Hospital A / Hospital B / Hospital C → local training → model updates → aggregation → global model.

Suggested Project Structure

```
FedMed/  
■■■ data/  
■■■ centralized/  
■ ■■■ dataset.py  
■ ■■■ model.py  
■ ■■■ train.py  
■ ■■■ evaluate.py  
■■■ federated/  
■ ■■■ server.py  
■ ■■■ client.py  
■ ■■■ config.py  
■■■ requirements.txt  
■■■ README.md
```

Initial Experiment Settings

Parameter	Starting Point
Dataset	Small BraTS subset
Model	MONAI 3D U-Net
Batch size	1
Learning rate	1e-4
Epochs	5 initially
Primary metric	Dice score
Federated clients	3 mock hospitals
Training style	First centralized, then federated

Important Scope Rule

Do not add TenSEAL, homomorphic encryption, secure MPC, or the React dashboard during Week 1. First prove that centralized training and ordinary federated training work. Privacy and dashboard features can be layered on afterward.

Week 1 Success Criteria

- A BraTS MRI volume and segmentation mask can be loaded successfully.
- The 3D U-Net can perform a forward pass.
- The model can train for multiple epochs.
- A validation Dice score is recorded as the centralized baseline.
- A Flower server runs successfully.
- Three independent mock hospital clients connect successfully.

- Each hospital can use a different local data partition.
- At least one federated round completes and produces an updated global model.

Key Principle

Week 1 is about establishing a reliable baseline and proving the federated infrastructure. The centralized model gives you a benchmark; the three hospital nodes give you the foundation for the privacy-preserving system that will be built in later weeks.